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Original Article Title: “Landfill Site Allocation Multi-Objective Optimization with NSGA-II: A Case Study on Fogo Island, Cape Verde”

To: IEEE Access Editor

Re: Response to reviewers

Dear Editor and Reviewers,

We sincerely thank you for the continued and constructive evaluation of our manuscript. We are grateful that Reviewers 1 and 3 found the previous revision satisfactory and raised no further concerns. We have carefully addressed all seven comments raised by Reviewer 2, and we believe this revised version fully resolves the remaining issues.

The key enhancements in this second revision include: (1) a new Table 4 explicitly documenting the normalization direction and objective assignment for each spatial criterion, including raw variable, desired direction of contribution, normalization formula, objective assignment, and optimization sign, directly addressing the reviewer's request for objective function transparency; (2) a moderated justification of the $k=3$ cardinality constraint, now presented as the most appropriate planning scenario within the identified operational constraints rather than a proven mathematical optimum, with the three supporting lines of evidence clearly separated; (3) a reconciled territorial interpretation of the selected solution, explicitly clarifying that the model optimizes declared spatial objectives and constraints rather than administrative coverage, with guidance for researchers requiring balanced municipal representation; (4) moderated operational claims throughout the Results and Conclusion sections, ensuring all statements remain within the scope of a strategic spatial siting framework and qualify operational conclusions with appropriate caveats; (5) refined exhaustive-search validation reporting, now including the True Pareto Front size ($|P^*| = 213$ solutions), the Generational Distance ($GD = 0.003558$), and the Inverted Generational Distance ($IGD = 0.003975$), and explicitly avoiding claims of full mathematical equivalence; (6) a precise definition of the Social Safety objective as the spatial separation from populated areas, aviation infrastructure, and agricultural land uses, collectively representing zones of heightened social exposure and land-use conflict; and (7) revised conclusions replacing overstatements about mathematical equivalence and operational robustness with calibrated language consistent with the evidence produced by the model.

All core results remain unchanged and continue to be validated against a complete deterministic Exhaustive Search over the 2.90 million feasible configurations of the constrained combinatorial space, demonstrating that NSGA-II achieves a high-quality approximation of the true Pareto front, capturing 99.50% of the true hypervolume in the best run (mean: 97.55% across 30 independent runs), with a 4.0-fold reduction in computational time relative to exhaustive enumeration.

We are submitting: (a) this point-by-point response document; (b) a highlighted revised manuscript (PDF); (c) a clean final manuscript (LaTeX + PDF); and (d) the public data and code repository (Zenodo/GitHub).

We believe the revised manuscript fully addresses all reviewer concerns and meets IEEE Access publication standards.

Respectfully submitted on behalf of all authors,

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Reviewer #1, Concern #1 (This paper can be accepted.)

Author response:

We sincerely thank Reviewer 1 for the thorough evaluation of our manuscript and for the positive assessment. We are pleased that the revisions addressing the concerns raised in the previous round were found satisfactory. The constructive feedback provided in the first round of review led to substantial improvements in the manuscript, and we are grateful for the reviewer's careful and detailed engagement with our work.

Reviewer #3, Concern #1 (The authors have carefully revised. I have no further comments.)

Author response:

We sincerely thank Reviewer 3 for the careful re-evaluation of the revised manuscript and for confirming that the previous revisions were thorough and satisfactory. The reviewer's constructive engagement throughout the review process contributed meaningfully to the quality of this work, and we are grateful for the time and expertise dedicated to evaluating our submission.

Reviewer #2, Concern #1 (Clarify the orientation of the objective functions): The manuscript defines suitability scores in which higher values generally indicate better conditions, but the Cost and Environmental Impact objectives are minimized. This is not necessarily wrong, but the transformation logic must be made explicit. For each criterion, the authors should specify whether the raw variable represents suitability, risk, cost, distance, or exposure; whether it is normalized directly or inversely; and how it contributes to f1, f2, or f3. A compact table listing raw variable, desired direction, normalization formula, objective assignment, and optimization sign would resolve this issue.

Author response:

We thank the reviewer for this precise and actionable request. We fully agree that making the normalization logic explicit for each criterion is essential for methodological transparency and reproducibility. The revised manuscript now includes a dedicated table and an accompanying explanatory paragraph that address every dimension of this concern.

We added Table 4, "Spatial Criteria: Normalization Direction and Objective Assignment", to Section II-C of the revised manuscript, immediately after the feature matrix paragraph and before the AHP subsection. The table specifies, for each of the eleven spatial criteria: (i) the criterion name and unit of measurement; (ii) the effect of increasing the raw variable value, providing physical interpretation; (iii) the normalization type applied (Direct, Inverse, or Binary); and (iv) the objective assignment and optimization sign.

The three normalization types are formally defined in the table footnotes. Direct normalization ($\hat{c} = (v - v_{\min}) / (v_{\max} - v_{\min})$) is applied to criteria where higher raw values contribute positively to the objective, such as greater distance from roads reducing infrastructure costs (f1) or greater distance from urban areas increasing the social buffer (f3). Inverse normalization ($\hat{c} = (v_{\max} - v) / (v_{\max} - v_{\min})$) is applied to criteria where shorter distances increase environmental risk, so that proximity to sensitive features such as protected

zones, rivers, coastlines, and wells produces larger normalized values and thus a higher environmental impact score in f_2 . Binary classification is applied to the Aspect criterion, where leeward aspects (90° – 180° relative to the dominant NE Trade Winds) receive score 0 (favorable), while all windward-facing aspects receive score 1 (penalized), following the regional environmental framework of Andrade and Barbosa [23].

An explanatory paragraph was also added immediately after the table, walking through the logic of each objective function explicitly so that readers can verify the consistency between the normalization direction and the minimization or maximization sign of each objective.

Author action:

We added Table 4, "Spatial Criteria: Normalization Direction and Objective Assignment", to Section II-C of the revised manuscript, along with a full explanatory paragraph. The table documents all eleven criteria with their unit, effect of increasing value, normalization type (Direct/Inverse/Binary), objective assignment (f_1 , f_2 , or f_3), and optimization sign (min/max). The footnotes formally define the Direct and Inverse normalization formulas and document the Binary aspect classification adapted from [23]. No other sections of the manuscript required modification to address this concern, as the normalization logic is now fully captured in this new table and its accompanying text.

Reviewer #2, Concern #2 (Strengthen the justification of $k = 3$): The manuscript now provides a more developed rationale for fixing the number of facilities at three. However, the justification still mixes algorithmic, administrative, territorial, and budgetary arguments. This section would be stronger if the authors clearly separated: (i) $k = 3$ as a planning assumption; (ii) sensitivity results across alternative k values; and (iii) operational arguments specific to Fogo Island. If the intention is to claim that $k = 3$ is operationally optimal, the supporting evidence should be stated more quantitatively. If not, the wording should be moderated to present $k = 3$ as a justified scenario rather than a proven optimum.

Author response:

We thank the reviewer for this precise and constructive observation. We agree that the previous version mixed distinct lines of evidence without clearly separating them, which weakened the overall justification and could suggest an overclaim regarding the optimality of $k = 3$. The revised manuscript now restructures this justification along three clearly separated lines of evidence, as recommended, and moderates the concluding language to present $k = 3$ as the most appropriate planning scenario within the identified operational constraints rather than a mathematically proven global optimum.

Regarding the three lines of evidence now explicitly separated in Section II-E:

The first line of evidence treats $k = 3$ as a planning assumption grounded in the geomorphological and administrative structure of Fogo Island. The island is divided into three principal municipal sectors (São Filipe, Mosteiros, and Santa Catarina), each with distinct settlement patterns and separated by a peripheral road network imposed by the central volcanic cone rising to 2,829 meters. This structure makes $k = 3$ the natural upper bound for a decentralized configuration that ensures independent service continuity across all sectors in the event of road network disruption. This argument is presented at the outset of the justification section before any algorithmic evidence is introduced, establishing $k = 3$ as a problem-defining parameter before it is evaluated as an optimization outcome.

The second line of evidence presents the sensitivity analysis results across k in the set $\{2, 3, 4, 5, 6, 7, 8, 10\}$. From an algorithmic perspective, the normalized hypervolume peaks at $k = 3$, which constitutes the elbow point of the performance curve. The sensitivity analysis further shows that $k = 3$ achieves a 249%

improvement in hypervolume over $k = 2$ (0.022 versus 0.006) and generates 59% more Pareto-optimal solutions (150 versus 95). These results are presented separately from the operational arguments, in Table 8 and Figure 3 of the revised manuscript, so that the algorithmic evidence stands on its own merits before being combined with the practical constraints.

The third line of evidence presents the operational arguments specific to Fogo Island, including municipal budget limitations, administrative capacity constraints, waste generation scale, and alignment with regional planning policy, all of which define an operationally feasible range of k equal to or less than 3. These arguments are drawn from documented sources specific to Small Island Developing States and to Cape Verde's institutional context, and they are presented as hard constraints that independently bound the feasible range, independently of the algorithmic analysis.

The concluding paragraph of Section II-E was specifically revised to moderate the language and to make the three-part structure explicit. The previous version stated that $k = 3$ "emerges as the optimal configuration within the operational constraints." The revised version now reads:

"Therefore, within the identified operational constraints, $k = 3$ emerges as the most appropriate planning scenario, representing the maximum viable investment in waste management infrastructure for Fogo Island. While $k = 3$ is not claimed as a mathematically proven global optimum across all possible contexts, the convergence of algorithmic evidence, sensitivity analysis, and documented SIDS-specific resource limitations provides a well-grounded justification for fixing the cardinality at this value."

This revised language directly implements the reviewer's recommendation to present $k = 3$ as a justified scenario rather than a proven optimum, while preserving the quantitative evidence that supports this choice.

Author action:

We revised the concluding paragraph of Section II-E to separate the three lines of evidence and to moderate the language as recommended. The key modification replaces the previous claim that $k = 3$ is "the optimal configuration" with the following passage, now present in the revised manuscript:

"Therefore, within the identified operational constraints, $k = 3$ emerges as the most appropriate planning scenario, representing the maximum viable investment in waste management infrastructure for Fogo Island. While $k = 3$ is not claimed as a mathematically proven global optimum across all possible contexts, the convergence of algorithmic evidence, sensitivity analysis, and documented SIDS-specific resource limitations provides a well-grounded justification for fixing the cardinality at this value. The results presented in the following sections focus on this operationally-optimal configuration."

The structural separation of the three arguments, namely (i) the planning assumption rooted in the island's geomorphological and administrative structure, (ii) the sensitivity analysis results across k values, and (iii) the operational constraints specific to Fogo Island, was achieved by reorganizing the existing content of Section II-E into three clearly delimited paragraphs, each introduced with a topic sentence identifying its specific line of reasoning. No new figures or tables were required to implement this change, as the existing Table 8 and Figure 3 already provide the quantitative sensitivity evidence referred to in the second line of argument.

Reviewer #2, Concern #3 (Reconcile the territorial interpretation of the selected solution): The final solution is discussed as providing territorial coverage and operational redundancy. However, the selected configuration does not appear to distribute one facility across each principal municipal sector. This does not invalidate the optimization result, but it requires a more careful interpretation. The authors should clarify that the model optimizes the stated spatial objectives and constraints, not necessarily

administrative coverage. If municipal representation is required, it should be added as an explicit constraint; otherwise, claims about municipal coverage should be softened.

Author response:

We thank the reviewer for this careful and fair observation. We fully agree that the previous version of the manuscript discussed the territorial distribution of the Best Compromise Solution in a way that implied broader municipal coverage than the configuration actually achieves. The reviewer is correct that the selected configuration, comprising sites 184, 430, and 460, assigns two facilities to the Mosteiros municipality and one to Santa Catarina, and does not include a facility in São Filipe. This distribution is a direct and legitimate result of the optimization pressure rather than a design failure, but it needed to be interpreted more carefully and honestly in the manuscript.

The revised manuscript addresses this concern in Section III-F through two specific changes.

First, the description of the spatial coverage was made precise. The previous version stated that "the three facilities cover Mosteiros and Santa Catarina" without specifying the within-municipality distribution. The revised version now reads:

"Together, the three facilities cover Mosteiros and Santa Catarina, with two sites assigned to Mosteiros and one to Santa Catarina, forming a geodesic triangle with an area of 52.82 km² and an average inter-facility spread of 13.41 km."

This formulation accurately reflects the actual distribution produced by the optimization model and avoids any implicit suggestion of one-facility-per-municipality coverage.

Second, a new paragraph was added to explicitly clarify the scope of the model and to provide guidance for researchers who require balanced administrative representation. The added text reads:

"It is important to note that the model optimizes the declared spatial objectives and constraints, not administrative coverage per se. The absence of a facility in São Filipe is therefore a result of the optimization pressure rather than a design failure. Researchers requiring balanced representation across all three municipalities should incorporate an explicit administrative coverage constraint into the formulation. The concentration in Mosteiros reflects the combined effect of the geographic dispersion constraint and the high Social Safety penalties associated with São Filipe's urban center, the largest on the island, which significantly reduces buffer distances from populated areas, the criterion with the highest AHP weight in f3 ($w = 0.166$, Table 7)."

This paragraph directly implements the reviewer's recommendation by clarifying that the model optimizes spatial objectives rather than administrative coverage, by reframing the absence of a facility in São Filipe as an optimization outcome rather than a deficiency, and by explicitly guiding future researchers on how to incorporate administrative coverage as a hard constraint if that property is required.

The existing quantitative explanation of why São Filipe is excluded, namely the high Social Safety penalties associated with the island's largest urban center, was retained and reinforced within this new paragraph to ensure the interpretation is grounded in the model's actual mechanics rather than in vague geographic reasoning.

Author action:

We made two targeted modifications to Section III-F of the revised manuscript. The first modification specifies the precise within-municipality distribution of the selected facilities by replacing the previous formulation with the following text, now present in the revised manuscript:

"Together, the three facilities cover Mosteiros and Santa Catarina, with two sites assigned to Mosteiros and one to Santa Catarina, forming a geodesic triangle with an area of 52.82 km² and an average inter-facility spread of 13.41 km."

The second modification adds a new paragraph immediately following the coverage description, explicitly clarifying the scope of the model and providing guidance for researchers requiring balanced administrative coverage. The full text of this addition, now present in Section III-F of the revised manuscript, reads:

"It is important to note that the model optimizes the declared spatial objectives and constraints, not administrative coverage per se. The absence of a facility in São Filipe is therefore a result of the optimization pressure rather than a design failure. Researchers requiring balanced representation across all three municipalities should incorporate an explicit administrative coverage constraint into the formulation. The concentration in Mosteiros reflects the combined effect of the geographic dispersion constraint and the high Social Safety penalties associated with São Filipe's urban center, the largest on the island, which significantly reduces buffer distances from populated areas, the criterion with the highest AHP weight in f3 ($w = 0.166$, Table 7)."

No modifications to other sections were required to address this concern.

Reviewer #2, Concern #4 (Moderate operational claims): The model is best understood as a strategic spatial siting framework. It does not explicitly model facility capacity, landfill lifespan, waste generation by service area, vehicle routing, fleet composition, or network-based transport cost. The manuscript acknowledges these limitations, but some conclusions still imply operational resilience and service continuity. These statements should be revised. The framework can support strategic decision making and identify spatially balanced candidate configurations, but it cannot by itself demonstrate full operational feasibility.

Author response:

We thank the reviewer for this precise and well-founded observation. We fully agree that the scope of the proposed framework is strategic spatial siting rather than operational logistics, and that certain passages in the previous version overstated the operational implications of the results. Following a comprehensive review of the full manuscript, we identified and corrected six passages that contained language inconsistent with the strategic scope of the model. Each modification is described below.

The first modification addresses the Figure 1 caption, where the previous version stated that the constraints were incorporated "to ensure operational resilience and territorial coverage." This phrasing implied that the model itself guarantees operational resilience, which goes beyond its validated scope. The revised caption now reads:

"constraints to support geographic resilience and territorial coverage."

The second modification addresses the Introduction, where the rationale for the three-site configuration previously stated that "the remaining facilities can maintain waste management services for the island population." This implied a guarantee of service continuity that the model cannot provide. The revised text now reads:

"suggesting that, from a strategic siting perspective, if one facility becomes inaccessible due to road blockages (landslides, volcanic activity), the remaining facilities are positioned to support continued waste management operations."

The third modification addresses the opening sentence of the operational resilience paragraph in Section III-F. The previous version stated that "the multi-site configuration provides essential operational resilience." The revised version now reads:

"the multi-site configuration is spatially designed to support operational resilience, a strategic priority for waste management in Small Island Developing States, subject to logistical verification beyond the scope of this framework."

The fourth modification addresses the claim about service continuity later in the same paragraph of Section III-F. The previous version stated that "the remaining facilities can maintain waste management services." The revised version now reads:

"this spatial redundancy suggests that, from a strategic siting perspective, if one facility becomes temporarily inaccessible due to natural hazards, the remaining facilities are positioned to support continued waste management operations, subject to logistical and operational verification beyond the scope of this framework."

The fifth modification addresses the concluding statement of Section III-F. The previous phrase "robust engineering solution" implies operational validation the model does not provide. The revised version now reads:

"demonstrating that the evolutionary framework delivers a validated spatial decision-support configuration that is spatially resilient within the modeled constraints and adaptable to the territory's physical characteristics."

The sixth modification addresses the Conclusion section, where the previous version stated "ensuring operational resilience against the landslides and volcanic activities that characterize the island." This phrase implied definitive operational robustness, which is inconsistent with the scope of the study. The revised version now reads:

"supporting spatial resilience against the landslides and volcanic activities that characterize the island."

All six modifications are consistent with the study's limitations, which are explicitly documented in the Conclusion section, where the absence of capacity constraints, landfill lifespan projections, and vehicle routing cost modeling is acknowledged as a consequence of data unavailability in the Small Island Developing States context rather than a methodological oversight.

Author action:

Following a comprehensive review of the full manuscript, we implemented six targeted modifications to ensure that all language related to operational resilience and service continuity accurately reflects the strategic scope of the framework.

The revised texts, now present in the manuscript, are as follows. In the Figure 1 caption: "constraints to support geographic resilience and territorial coverage." In the Introduction: "the remaining facilities are positioned to support continued waste management operations." In Section III-F, opening of the resilience paragraph: "the multi-site configuration is spatially designed to support operational resilience, a strategic

priority for waste management in Small Island Developing States, subject to logistical verification beyond the scope of this framework." In Section III-F, service continuity sentence: "this spatial redundancy suggests that, from a strategic siting perspective, if one facility becomes temporarily inaccessible due to natural hazards, the remaining facilities are positioned to support continued waste management operations, subject to logistical and operational verification beyond the scope of this framework." At the end of Section III-F: "demonstrating that the evolutionary framework delivers a validated spatial decision-support configuration that is spatially resilient within the modeled constraints and adaptable to the territory's physical characteristics." In the Conclusion, third paragraph: "supporting spatial resilience against the landslides and volcanic activities that characterize the island."

These six modifications collectively ensure that the manuscript consistently presents the framework as a validated spatial decision-support tool, without implying operational service guarantees or engineering validation beyond the modeled spatial constraints.

Reviewer #2, Concern #5 (Refine the interpretation of exhaustive-search validation): The exhaustive-search comparison is a valuable strength of the paper. However, the manuscript should avoid stating or implying full mathematical equivalence unless the complete NSGA-II Pareto set is shown to coincide with the true Pareto front. Capturing 99.50% of the true-front hypervolume in the best run is strong evidence of high-quality approximation, but it is not identical to exact front recovery. The authors should report the size of the true Pareto front, the overlap between the heuristic and exhaustive fronts, and any additional metrics already introduced in the methodology, such as GD and IGD.

Author response:

We thank the reviewer for this rigorous and precise observation. We fully agree that the previous version of the manuscript overstated the validation results by using the phrase "proving that the heuristic approximation is mathematically equivalent to the true global limit," which implies exact Pareto front coincidence rather than a high-quality approximation. Capturing 99.50% of the true-front hypervolume in the best run is indeed strong evidence of approximation quality, but it does not constitute full mathematical equivalence. Furthermore, the Generational Distance (GD) and Inverted Generational Distance (IGD) metrics were formally defined in the methodology section but their numerical values were never reported in the results section, representing an inconsistency that the reviewer correctly identified. All three issues have been resolved in the revised manuscript.

Three specific changes were made to Section III-D.

The first change corrects the overstatement of mathematical equivalence. The previous version stated:

"confirming that the heuristic Pareto front contains all three extreme solutions, proving that the heuristic approximation is mathematically equivalent to the true global limit."

The revised version now reads:

"confirming that the heuristic Pareto front contains all three extreme solutions and constitutes a high-quality approximation of the true Pareto front, achieving 99.50% hypervolume capture in the best run (mean: 97.55% across 30 independent runs)."

This revised formulation accurately characterizes the result as a high-quality approximation supported by quantitative evidence, without asserting exact front coincidence.

The second change reports the size of the true Pareto front and the computed values of GD and IGD, which were defined in the methodology but never reported in the results. A new paragraph was added to Section III-D immediately after the validation table, with the following text now present in the revised manuscript:

"To further characterize the quality of the heuristic approximation, four complementary metrics were computed against the True Pareto Front ($|P^*| = 213$ solutions identified through exhaustive enumeration). The Optimality Gap $\delta_m = 0.0000$ for all three objectives confirms that NSGA-II recovered the exact extreme solutions. The Generational Distance $GD(A, P^*) = 0.003558$ indicates high convergence accuracy, while the Inverted Generational Distance $IGD(A, P^*) = 0.003975$ confirms comprehensive coverage of the trade-off surface without significant gaps. These results collectively demonstrate that NSGA-II achieves a high-quality approximation of the true Pareto front, capturing 99.50% of its hypervolume in the best run (mean: 97.55% across 30 independent runs), rather than claiming full mathematical equivalence with the exhaustive front."

This paragraph directly implements the reviewer's request by reporting the true Pareto front size ($|P^*| = 213$), the GD and IGD values that were already defined in the methodology, and by explicitly qualifying the result as a high-quality approximation rather than an exact recovery.

The third change adds a footnote to the validation table (Table 13) to clarify that the NSGA-II Best column reports the independent extreme solutions of the Pareto front rather than a single consolidated solution, preventing the misreading that a single solution simultaneously achieves all three extreme values. The footnote now reads:

"Values represent the independent extreme solutions of the Pareto front, not a single consolidated solution."

Author action:

We made three targeted modifications to Section III-D of the revised manuscript.

The first modification replaces the phrase "mathematically equivalent to the true global limit" with a quantitatively grounded characterization of approximation quality. The revised text, now present in the manuscript, reads:

"confirming that the heuristic Pareto front contains all three extreme solutions and constitutes a high-quality approximation of the true Pareto front, achieving 99.50% hypervolume capture in the best run (mean: 97.55% across 30 independent runs)."

The second modification adds a new paragraph after the validation table reporting the true Pareto front size, GD, and IGD values. The full text of this addition, now present in Section III-D of the revised manuscript, reads:

"To further characterize the quality of the heuristic approximation, four complementary metrics were computed against the True Pareto Front ($|P^*| = 213$ solutions identified through exhaustive enumeration). The Optimality Gap $\delta_m = 0.0000$ for all three objectives confirms that NSGA-II recovered the exact extreme solutions. The Generational Distance $GD(A, P^*) = 0.003558$ indicates high convergence accuracy, while the Inverted Generational Distance $IGD(A, P^*) = 0.003975$ confirms comprehensive coverage of the trade-off surface without significant gaps. These results collectively demonstrate that NSGA-II achieves a high-quality approximation of the true Pareto front, capturing 99.50% of its hypervolume in the best run (mean: 97.55% across 30 independent runs), rather than claiming full mathematical equivalence with the exhaustive front."

The third modification adds a clarifying footnote to Table 13 (Algorithmic Validation). The footnote reads:

"Values represent the independent extreme solutions of the Pareto front, not a single consolidated solution."

No modifications to other sections were required to address this concern.

Reviewer #2, Concern #6 (Tighten terminology around "Social Safety"): The third objective is based mainly on spatial separation from urban areas, aerodrome infrastructure, and cultivated areas. This is a valid planning dimension, but the term "Social Safety" may be broader than what the variables measure. The authors should either define the term more narrowly or rename it to something closer to the implemented construct.

Author response:

We thank the reviewer for this precise terminological observation. We agree that the previous version of the manuscript introduced the term "Social Safety" without providing a sufficiently narrow definition that would anchor it to the specific variables implemented in the optimization model. Without this definition, the term could be interpreted more broadly than the model actually captures, which may create a gap between the reader's expectations and the implemented construct.

After careful consideration, we decided to retain the term "Social Safety" rather than rename it, for two reasons. First, the term is already established throughout the manuscript, including in multiple tables, figures, equations, and narrative sections, and renaming it at this stage would require extensive revisions that could introduce inconsistencies. Second, the term is defensible within the scope of the model when properly defined, as it captures the spatial separation from areas of heightened social exposure, including populated zones, aviation infrastructure, and agricultural land uses, which together represent the primary land-use conflict zones relevant to landfill siting in this context.

To resolve the concern, we added a precise, narrow definition of Social Safety directly within the problem formulation in Section II-D, immediately at the point where f_3 is introduced. The previous version stated only that f_3 "represents Social Safety and is maximized," without specifying what Social Safety means within the model. The revised version now reads:

"while $f_3(x)$ represents Social Safety, defined here as the spatial separation from populated areas, aviation infrastructure, and agricultural land uses, collectively representing zones of heightened social exposure and land-use conflict, and is maximized."

This definition explicitly bounds the construct to the three criteria that constitute f_3 in the model, namely distance to urban areas, distance to aerodrome, and distance to cultivated areas, and characterizes them as indicators of social exposure and land-use conflict rather than as a general measure of public safety or well-being. This formulation makes it clear to readers that Social Safety in this study refers specifically and exclusively to spatial isolation from these three categories of sensitive receptors, preventing any broader interpretation of the term.

Author action:

We made one targeted modification to Section II-D of the revised manuscript, adding a precise definition of the Social Safety objective at the point of its formal introduction. The previous statement was:

"while $f_3(x)$ represents Social Safety and is maximized."

The revised text, now present in the manuscript, reads:

"while $f_3(x)$ represents Social Safety, defined here as the spatial separation from populated areas, aviation infrastructure, and agricultural land uses, collectively representing zones of heightened social exposure and land-use conflict, and is maximized."

This definition narrows the construct to the specific variables implemented in the model and grounds the terminology in the spatial planning literature on land-use conflict and social exposure, directly addressing the reviewer's concern without requiring renaming or extensive revision of other sections. No other modifications were required to address this concern.

Reviewer #2, Concern #7 (Reduce overstatement in the conclusions): The conclusions should be revised to ensure that all claims remain within the evidence produced by the model. Phrases implying definitive operational robustness, mathematical equivalence, or complete system-level feasibility should be replaced with more precise language, such as "within the modeled spatial constraints," "high-quality approximation," "strategic screening," and "decision-support framework."

Author response:

We thank the reviewer for this important and fair observation. We agree that the previous version of the Conclusion contained two specific statements that exceeded the evidence produced by the model. The first overstated the validation result by claiming mathematical equivalence with the true global limit. The second framed the computational speedup as proof of scalability rather than as supporting evidence. Both statements have been revised in the updated manuscript to use precise language that accurately reflects the scope and evidence of the study.

The first overstatement appeared in the second paragraph of the Conclusion, where the previous version stated:

"confirming that the NSGA-II Pareto front contains all three extreme solutions and is mathematically equivalent to the true global limit."

As noted in our response to Concern 5, capturing 99.50% of the true-front hypervolume in the best run is strong evidence of approximation quality but does not constitute full mathematical equivalence, since the NSGA-II approximation set of 150 solutions does not coincide exactly with the true Pareto front of 213 solutions identified through exhaustive enumeration. The revised version now reads:

"confirming that the NSGA-II Pareto front contains all three extreme solutions and constitutes a high-quality approximation of the true Pareto front (99.50% hypervolume capture in the best run; mean: 97.55% across 30 independent runs)."

This formulation accurately characterizes the result as a high-quality approximation supported by quantitative evidence, without asserting exact coincidence with the true front.

The second overstatement appeared in the same paragraph, where the previous version stated:

"Furthermore, this performance was achieved with a 4.0-fold speedup over the brute-force baseline, proving the framework's scalability to larger territories where exhaustive enumeration is intractable."

The word "proving" is too strong in this context. The 4.0-fold speedup demonstrates computational advantage on this specific problem instance, but it does not constitute proof of general scalability, since the

exhaustive search baseline is itself specific to the feasibility conditions of this case study, as discussed in the scalability analysis of Section III-D. The revised version now reads:

"Furthermore, this performance was achieved with a 4.0-fold reduction in computational time relative to exhaustive enumeration at this problem scale, supporting the framework's scalability to larger territories where exhaustive enumeration is intractable."

The replacement of "proving" with "supporting" and the addition of the qualifier "at this problem scale" ensure that the claim is precisely bounded to what the evidence can support, consistent with the four-factor scalability analysis already presented in Section III-D.

These two revisions, combined with the moderation of operational claims in Section III-F addressed in Concern 4, ensure that the Conclusion as a whole uses language that is calibrated to the actual evidence produced by the model and that does not extend beyond the scope of a validated spatial decision-support framework.

Author action:

We made two targeted modifications to the Conclusion section of the revised manuscript.

The first modification replaces the phrase "mathematically equivalent to the true global limit" with a quantitatively grounded characterization of approximation quality. The revised text, now present in the Conclusion, reads:

"confirming that the NSGA-II Pareto front contains all three extreme solutions and constitutes a high-quality approximation of the true Pareto front (99.50% hypervolume capture in the best run; mean: 97.55% across 30 independent runs)."

The second modification replaces the word "proving" and the unqualified scalability claim with precise language bounded to the problem scale of this study. The revised text, now present in the Conclusion, reads:

"Furthermore, this performance was achieved with a 4.0-fold reduction in computational time relative to exhaustive enumeration at this problem scale, supporting the framework's scalability to larger territories where exhaustive enumeration is intractable."

No other modifications to the Conclusion were required to address this concern. The combined effect of these two revisions, together with the modifications to Section III-F introduced in response to Concern 4, ensures that the Conclusion as a whole is fully consistent with the evidence produced by the model and uses language appropriate to a validated spatial decision-support framework.